

PHYS 225

Week 11, Lecture 9:
Vector fields, curvilinear coordinates, and
vector differential operators



UNIVERSITY OF
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Objectives:

- Review scalar and vector fields
- Review definitions and interpretation of gradient, divergence, and curl
- Get (re)-acquainted with with curvilinear coordinates

One Minute Paper: In classical mechanics, is an object's acceleration always in the same direction as the net force on it?

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Scalar fields

- Field: some function of space (and time)
- Scalar field: function of space (and time) that returns a single number (in this class, a real number)

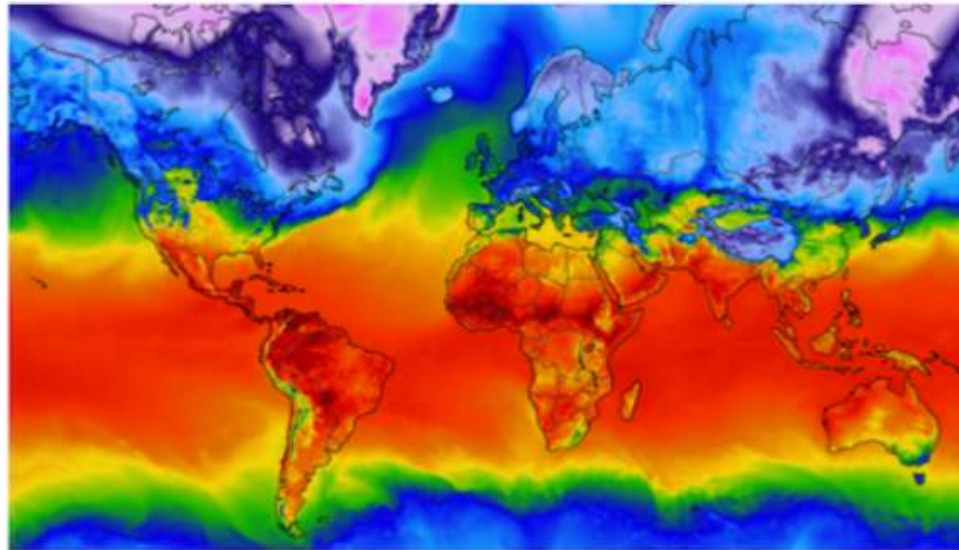
$$\phi : \mathbb{R}^n \rightarrow \mathbb{R}$$

- At every point $x^\mu \in \mathbb{R}^n$, there is some number $\phi(x^\mu)$
- Examples of scalar fields: temperature, mass density, electrostatic potential

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Scalar fields

Example: temperature



Temperature
map of world

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Vector fields

- Vector Field: function of space (and time) that returns a vector

$$\vec{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

- Attached to every point $x^\mu \in \mathbb{R}^n$ is an m -component vector

$$\vec{y} = \vec{F}(x^\mu)$$

- Examples: electric and magnetic fields, velocity

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Vector fields

Examples: wind on the surface of the earth



Wind map of
world

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Differentiating Fields

- We want to be able to differentiate our fields to see how they change as we move around
- Scalar and vector fields are different objects, so we need different ways to differentiate them
- Derivative of scalar field: **Gradient**
- Two different ways to differentiate a vector field:
 - **Divergence** $\iff$ Dot product
 - **Curl** $\iff$ Cross product (only in 3 dimensions!)

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Gradient

- Given Cartesian coordinates x^i with $i = 1, 2, 3$ on $\mathbb{R}^3$, the gradient is defined as:

$$\vec{\nabla} \phi = \frac{\partial \phi}{\partial x} \hat{x} + \frac{\partial \phi}{\partial y} \hat{y} + \frac{\partial \phi}{\partial z} \hat{z}$$

$\Rightarrow$ Differentiating a scalar field leaves us with a vector field!

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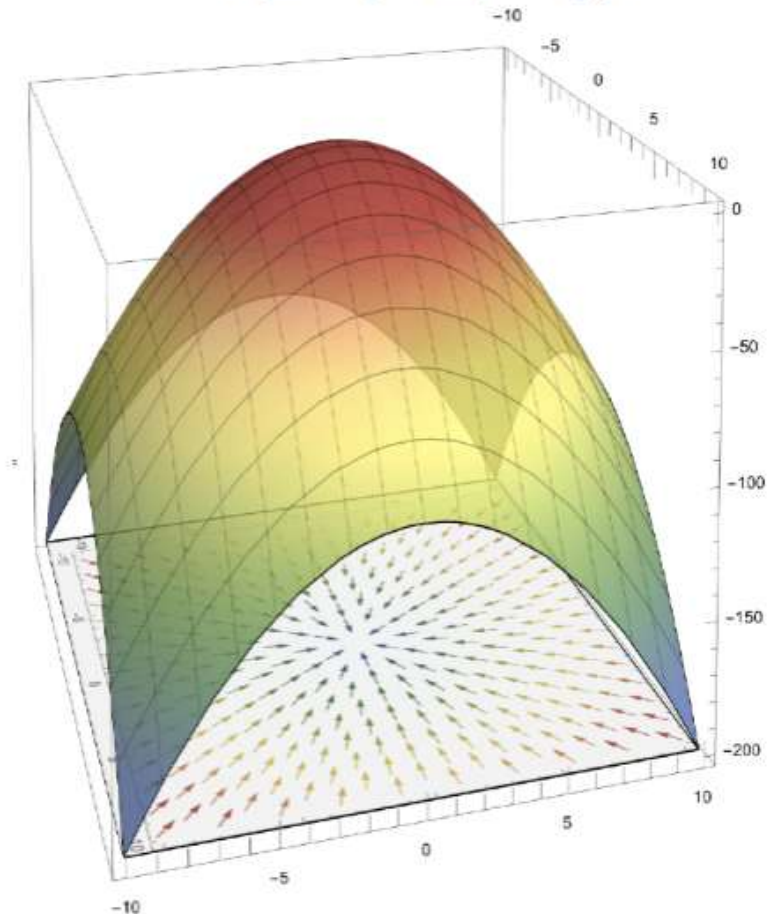
Gradient

What is the gradient actually doing?

- The gradient $\vec{\nabla} \phi$ points in the direction of maximum **increase**
- The magnitude $|\vec{\nabla} \phi|$ gives the rate of increase (slope) along this direction

$$\phi(x, y) = -x^2 - y^2$$

$$\vec{\nabla} \phi = (-2x, -2y)$$

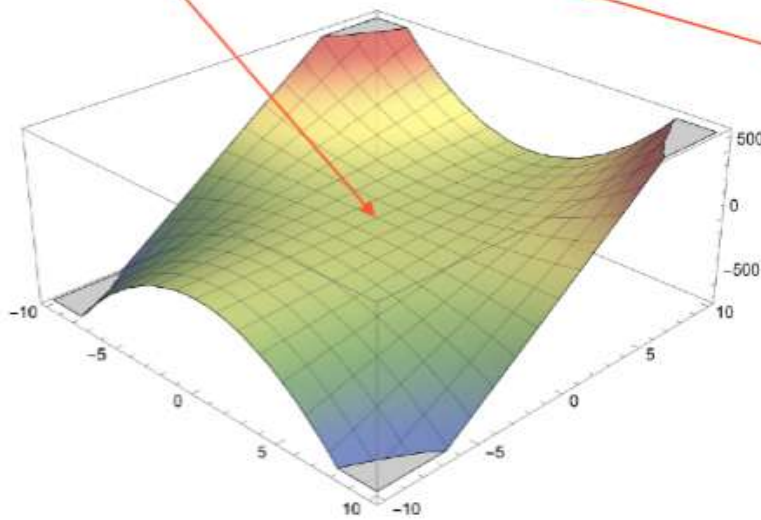


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Gradient

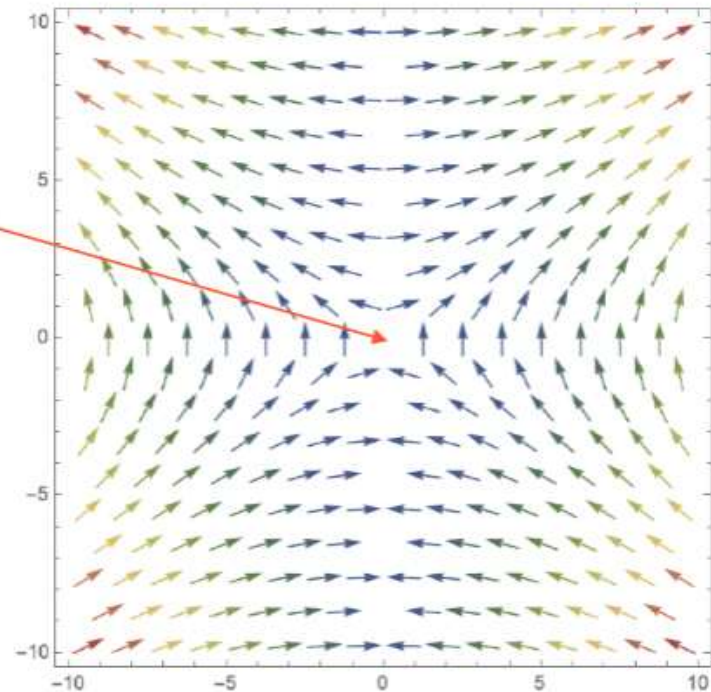
More examples!

Gradient vanishes at (0,0):
the field is approximately
flat there



$$\phi(x, y) = x^2y$$

$$\vec{\nabla} \phi = (2xy, x^2)$$



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Gradient

The gradient as an operator: $\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$
“del” or “nabla”

Note: the arrow on top reminds us this is a vector operator!

$\vec{\nabla}$ can eat a scalar field and spit out a vector:

$$\vec{\nabla} f = \left(\hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z} \right) f$$

(analogous to a matrix operating on a column vector)

In index notation: $\vec{\nabla} \Leftrightarrow \partial_i$ [Discussion 9]

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Divergence

Divergence: “dot product” between $\vec{\nabla}$ and a vector field $\vec{F}$

$$\text{In 3D: } \vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$\Rightarrow$ Divergence of a vector field is a scalar field

(same operator can eat a vector and return a number)

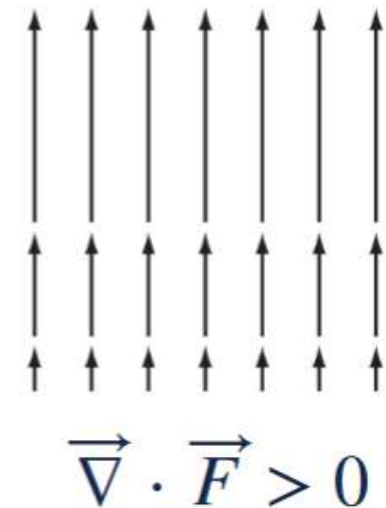
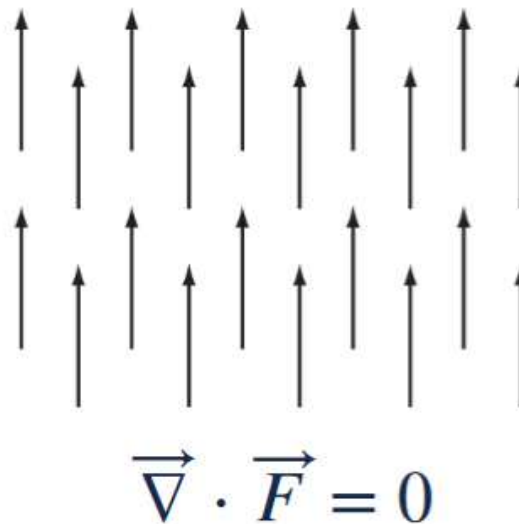
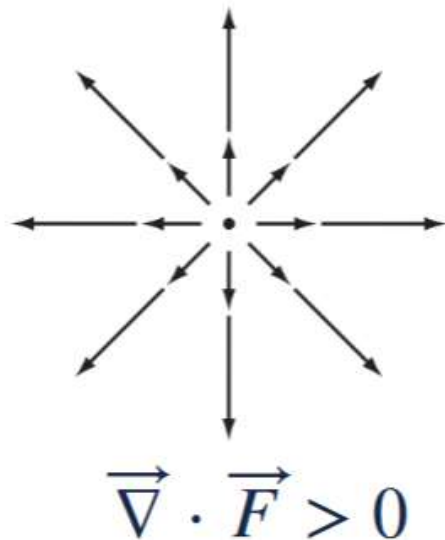
$$\text{In index notation: } \vec{\nabla} \cdot \vec{F} \Leftrightarrow \partial_i F^i \quad [\text{HW 9}]$$

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Divergence

What is the divergence actually doing?

Measures how much the vectors described by $\vec{F}$ “spread out” (or “diverge”) from a given point



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Curl

Curl: “cross product” between $\vec{\nabla}$ and a vector field $\vec{F}$

$\Rightarrow$ Curl of vector field gives another vector field

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}, \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}, \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

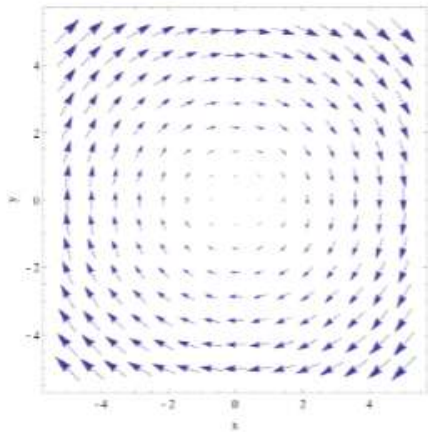
Note: while divergence can be defined in any dimension, the curl **only** makes sense for vector fields in 3D, $\vec{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$

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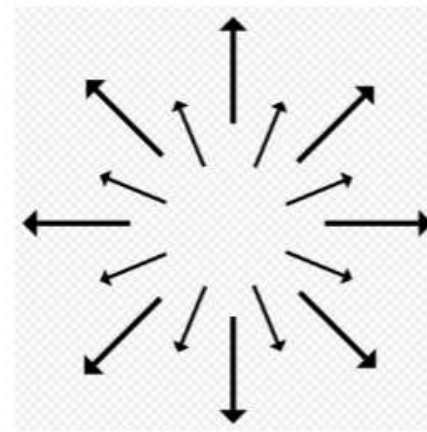
Curl

What is the curl actually doing?

Measures how much the vectors described by $\vec{F}$ circulate (or “curl”) around a point: imagine the motion of a paddlewheel



$$\vec{\nabla} \times \vec{F} \neq 0$$



$$\vec{\nabla} \times \vec{F} = 0$$

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Linearity

$\vec{\nabla}$ is a **linear** differential operator

- $\vec{\nabla}(\alpha\phi + \psi) = \alpha\vec{\nabla}\phi + \vec{\nabla}\psi$
- $\vec{\nabla} \cdot (\alpha\vec{F} + \vec{G}) = \alpha\vec{\nabla} \cdot \vec{F} + \vec{\nabla} \cdot \vec{G}$
- $\vec{\nabla} \times (\alpha\vec{F} + \vec{G}) = \alpha\vec{\nabla} \times \vec{F} + \vec{\nabla} \times \vec{G}$

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Leibniz Properties

Gradient, divergence, and curl obey product rules:

- $\vec{\nabla}(\phi\psi) = \phi \vec{\nabla}\psi + \psi \vec{\nabla}\phi$
- $\vec{\nabla} \cdot (\phi \vec{F}) = (\vec{\nabla}\phi) \cdot \vec{F} + \phi(\vec{\nabla} \cdot \vec{F})$
- $\vec{\nabla} \times (\phi \vec{F}) = (\vec{\nabla}\phi) \times \vec{F} + \phi(\vec{\nabla} \times \vec{F})$
- $\vec{\nabla} \cdot (\vec{F} \times \vec{G}) = (\vec{\nabla} \times \vec{F}) \cdot \vec{G} - \vec{F} \cdot (\vec{\nabla} \times \vec{G})$

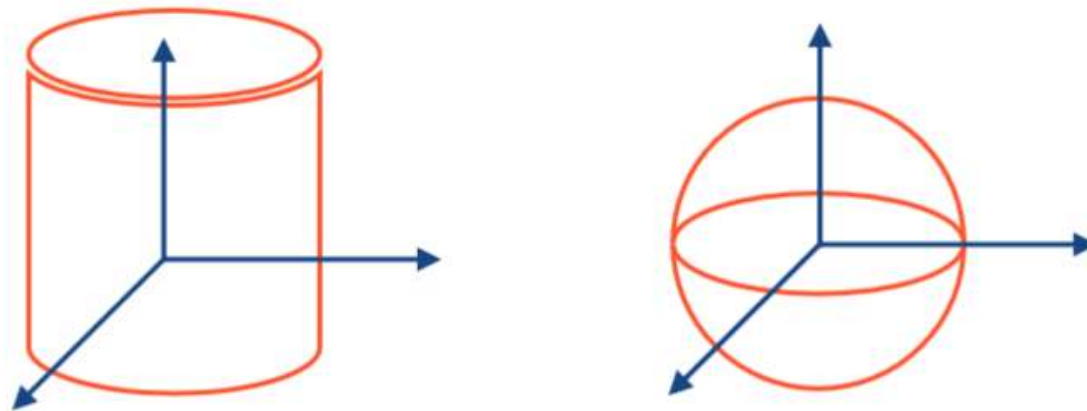
(no need to memorize these!!)

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Orthogonal Curvilinear Coordinates

We don't need to always use the Cartesian coordinate system.

(And it is often more convenient to not use it)



A useful generalization is an orthonormal coordinate system:

Unit vectors $\vec{e}_i$ satisfy $\vec{e}_i \cdot \vec{e}_i = 1$ and $\vec{e}_i \cdot \vec{e}_j = 0$

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Cylindrical Coordinates

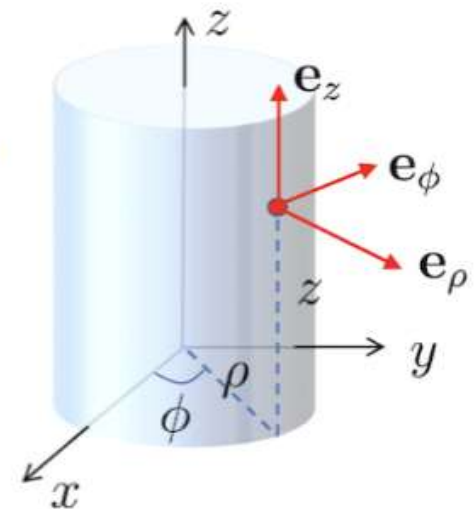
(ρ, ϕ, z) with $\rho \geq 0$, $0 \leq \phi < 2\pi$, $-\infty < z < \infty$

Cartesian coords in terms of cylindrical coords:

$$x = \rho \cos \phi \quad y = \rho \sin \phi \quad z = z$$

Cylindrical coords in terms of cartesian coords:

$$\rho = \sqrt{x^2 + y^2} \quad \tan \phi = \frac{y}{x}$$



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Spherical Coordinates

(r, θ, ϕ) with $r \geq 0$, $0 \leq \theta \leq \pi$, $0 \leq \phi < 2\pi$

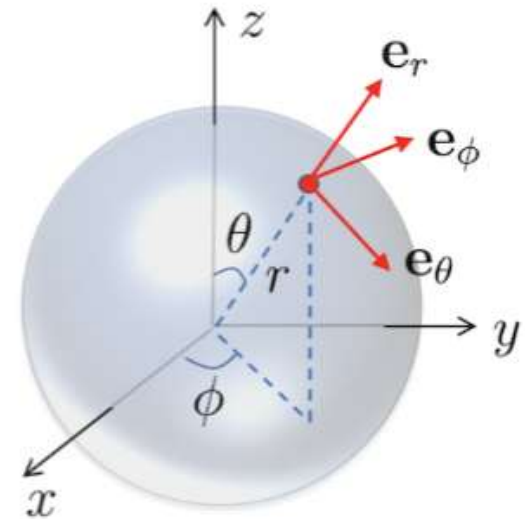
Cartesian coords in terms of spherical coords:

$$x = r \sin \theta \cos \phi \quad y = r \sin \theta \sin \phi \quad z = r \cos \theta$$

NOTE: physics uses θ for angle from z-axis ("polar angle"), while math uses ϕ . In this class, always physics notation!

Spherical coords in terms of cartesian coords:

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \tan \theta = \frac{\sqrt{x^2 + y^2}}{z}, \quad \tan \phi = \frac{y}{x}$$



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Grad/div/curl in cylindrical coordinates

The gradient, divergence, and curl are different in different coordinate systems:

Gradient: $\vec{\nabla} f = \frac{\partial f}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \hat{\phi} + \frac{\partial f}{\partial z} \hat{z}$ [Discussion 9]

Divergence: $\vec{\nabla} \cdot \vec{F} = \frac{1}{\rho} \frac{\partial(\rho F_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}$

Curl:

$$\vec{\nabla} \times \vec{F} = \left(\frac{1}{\rho} \frac{\partial F_z}{\partial \phi} - \frac{\partial F_\phi}{\partial z} \right) \hat{\rho} + \left(\frac{\partial F_\rho}{\partial z} - \frac{\partial F_z}{\partial \rho} \right) \hat{\phi} + \frac{1}{\rho} \left(\frac{\partial(\rho F_\phi)}{\partial \rho} - \frac{\partial F_\rho}{\partial \phi} \right) \hat{z}$$

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Grad/div/curl in spherical coordinates

The gradient, divergence, and curl are different in different coordinate systems:

Gradient: $\vec{\nabla} f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$ [Discussion 9]

Divergence: $\vec{\nabla} \cdot \vec{F} = \frac{1}{r^2} \frac{\partial(r^2 F_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta F_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}$

Curl:

$$\vec{\nabla} \times \vec{F} = \frac{1}{r \sin \theta} \left(\frac{\partial(\sin \theta F_\phi)}{\partial \theta} - \frac{\partial F_\theta}{\partial \phi} \right) \hat{r} + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial F_r}{\partial \phi} - \frac{\partial(r F_\phi)}{\partial r} \right) \hat{\theta} + \frac{1}{r} \left(\frac{\partial(r F_\theta)}{\partial r} - \frac{\partial F_r}{\partial \theta} \right) \hat{\phi}$$

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Tips for curvilinear coordinates

Unlike Cartesian coordinates, the spherical and cylindrical unit vectors change with position: they are true vector fields!

This means that the unit vectors can have nonzero divergence (**HW 9**).

If you are ever confused about a calculation, convert back to Cartesian coordinates.

(The formulas for div and curl are tricky to derive. **Don't** memorize them, they will be given on formula sheets.)

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Summary

- We can differentiate fields with the $\vec{\nabla}$ operator
 - Gradient (applied to scalar field, gives vector field)
 - Divergence (applied to vector field, gives scalar field)
 - Curl (applied to vector field, gives vector field)
- Can define orthogonal coordinate systems (e.g. cylindrical and spherical), which will be useful going forward

One Minute Paper: Besides relativistic force, what is another example of a vector field?